Neural Networks with Differentiable Belief-Function Layers

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Code: github.com/sulcantonin/enn-public

The claim

● Fusing evidence means deciding where the mass that sources *disagree* on goes.

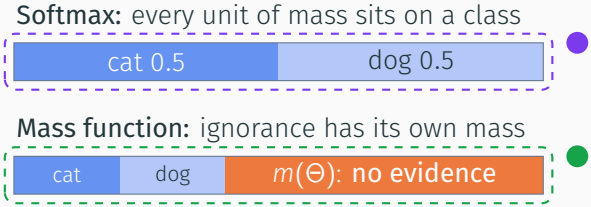
Yager (1987)
Conflict is ignorance. Send κ to Θ.

Back-propagation
Gradients must stay bounded where sources disagree.

● Same rule, same root: **no** $\frac{1}{1-\kappa}$ **normaliser**.

● Trained on hard labels only, the output $m(\Theta)$ orders *human* disagreement on images and on language.

Softmax has no slot for “I do not know”



Blurry cat-or-dog photo
Evidence for both. Softmax: (0.5, 0.5).

Fire hydrant
Evidence for neither. Softmax: (0.5, 0.5).

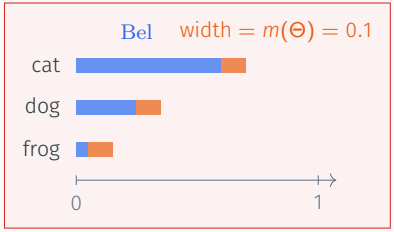
DST has always had the extra slot. ● Does it survive back-propagation?

A deliberately small focal set

$$\mathcal{F} = \big\{\{c_1\}, \ldots, \{c_K\},\ \Theta\big\}, \qquad |\mathcal{F}| = K{+}1 \ \text{ instead of } 2^K$$

- **Quasi-Bayesian** (Daniel 2012).
  Not a general belief function on 2^Θ.
- Exact, not a surrogate:

  $\mathrm{Bel}(\{c_i\}) = m(\{c_i\})$
  $\mathrm{Pl}(\{c_i\}) = m(\{c_i\}) + m(\Theta)$
- Every class has the same interval width, $m(\Theta)$.



Linear in K; duality $\mathrm{Pl}(A) = 1 - \mathrm{Bel}(\bar{A})$ holds exactly.

Two heads on $\mathcal{F}$: Dempster vs. Yager

Same intersection, different fate for the conflict $\kappa = \sum_{i \neq j} m_1(\{c_i\})\, m_2(\{c_j\})$:

$$m_{1 \boxplus 2}(\{c_i\}) = m_1(c_i)m_2(c_i) + m_1(c_i)m_2(\Theta) + m_1(\Theta)m_2(c_i)$$

$$m_{1 \boxplus 2}(\Theta) = \underbrace{m_1(\Theta)m_2(\Theta)}_{\text{nobody committed}} + \underbrace{\kappa}_{\text{committed, but disagreed}}$$

Dempster divides everything by $1 - \kappa$ and drops κ.

	cat	dog	Θ
head 1	0.60	0.10	0.30
head 2	0.10	0.60	0.30
Yager $\boxplus$	0.27	0.27	**0.46** $= 0.09 + 0.37$
Dempster $\oplus$	0.43	0.43	0.14

Back-propagation through Dempster's rule

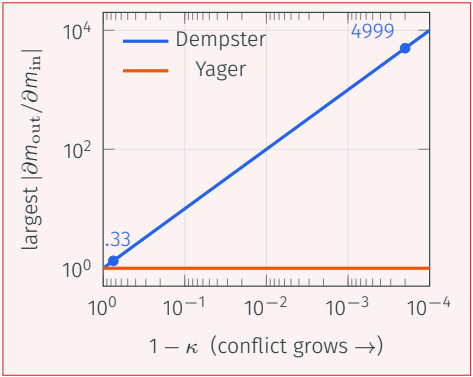
Quotient rule on $m_{1\oplus 2}(A) = \tilde{m}(A)/(1-\kappa)$:

$$\frac{\partial m_{1\oplus 2}(A)}{\partial m_1(B)} = \frac{1}{1-\kappa}\frac{\partial \tilde{m}(A)}{\partial m_1(B)} + \frac{\tilde{m}(A)}{(1-\kappa)^2}\frac{\partial \kappa}{\partial m_1(B)}$$

Head 1: p on c_1; head 2: p on c_2; rest on Θ ($\kappa = p^2$).

- **Dempster**: largest partial $= \dfrac{1}{1-\kappa}$
  1.33 at κ=0.25, **4999** at κ=0.9998
- Yager: 1.00 at every κ

Training Dempster needs a clamp on $1-\kappa$: a numerical guard with no belief-function reading.



Theorem 1: Yager cannot amplify a gradient

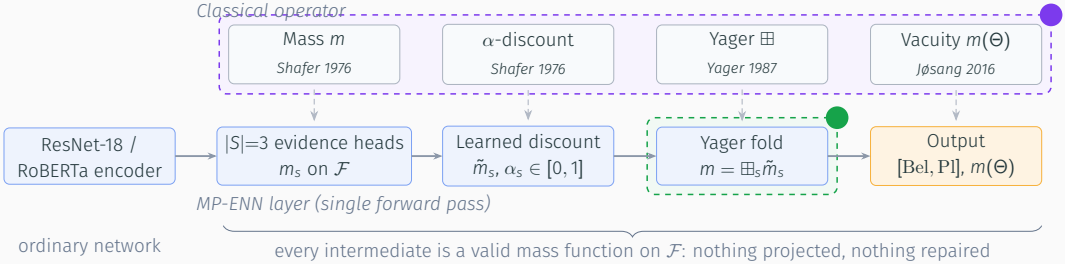
Theorem 1 (gradient stability of Yager)

For all $A, B \in \mathcal{F}$ and $i \in \{1, 2\}$: $\left|\partial m_{1 \boxplus 2}(A) / \partial m_i(B)\right| \leq 1$.
With one argument fixed, $m_1 \mapsto m_{1 \boxplus 2}$ is linear and column-stochastic, hence **1-Lipschitz in** ℓ_1 on the simplex.

Why. Fix m_2: each column of the Jacobian is m_2 redistributed over $\mathcal{F}$, so it is non-negative and sums to 1.

- **Elementary** as algebra. The point is the coincidence: the elementary property picks the rule chosen in 1987 for epistemic reasons.
- **Tight.** The two-head example attains 1.00.
- **Scope.** The combination layer. The encoder is an ordinary network.

MP-ENN: off-the-shelf DST operators, end to end



- The *same* $m(\Theta)$ is Jøsang's vacuity, the OOD score and the disagreement tracker: **one quantity, not three**.

The three equations

1. **Evidence → mass** (Sensoy's parametrisation read as a mass; Walley's IDM. Not new.)

$$e_{i,s} = \mathrm{softplus}(\mathbf{w}_{i,s}^{\top}\mathbf{h} + b_{i,s}), \qquad m_s(\{c_i\}) = \frac{e_{i,s}}{E_s + W}, \quad m_s(\Theta) = \frac{W}{E_s + W}, \quad W = K$$

2. **Discount, then fold in a fixed order** ($\alpha_1 = 1$; $\boxplus$ is commutative, not associative)

$$\tilde{m}_s(\{c_i\}) = \alpha_s m_s(\{c_i\}), \quad \alpha_s = \sigma\big(g_\phi[\mathbf{e}_s; \bar{\mathbf{e}}; \mathbf{e}_s - \bar{\mathbf{e}}]\big), \quad m = (\tilde{m}_1 \boxplus \tilde{m}_2) \boxplus \tilde{m}_3$$

3. **Loss on hard labels** (proper given $m(\Theta)$; not jointly proper on the $(K{+}1)$-simplex)

$$\mathcal{L} = -\log m(\{c_y\}) + \lambda_{\mathrm{unc}}\, m(\Theta) + \tfrac{\lambda_{\mathrm{aux}}}{|S|} \textstyle\sum_s -\log m_s(\{c_y\}), \qquad \lambda_{\mathrm{unc}} = \lambda_{\mathrm{aux}} = 0.1$$

Hard labels, yet vacuity tracks disagreement. Why?

- **Looks contradictory**
  - The loss sees one label per input.
  - $\lambda_{\text{unc}}\, m(\Theta)$ pushes vacuity *down*.
- **Is not**
  - λ_{unc} is uniform over inputs: it sets the *level*, not the *ordering*.
  - Ambiguous input $\Rightarrow$ heads disagree $\Rightarrow \kappa \uparrow$.

$$m(\Theta) = m_1(\Theta) m_2(\Theta) + \kappa$$

No loss term reaches κ directly.
Annotator distributions are never seen: any alignment is zero-shot.

Setup

Vision

- ResNet-18, CIFAR-10, 200 epochs, 10 methods $\times$ 5 seeds
- ● Grad-norm clip 1.0 for *all* methods; norms reported pre-clip
- ● Dempster: $1 - \kappa$ clamped at 10^{-8}

Language

- RoBERTa-base, SNLI (91.7% val.), W=3

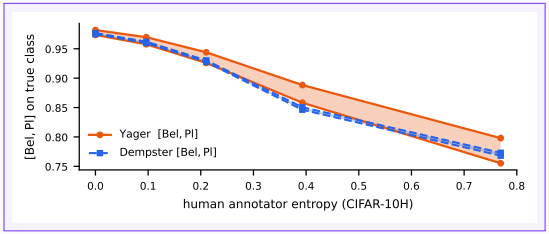
● **Human disagreement** (evaluation only)

- CIFAR-10H: 50 annotators / image
- ChaosNLI: 100 annotators / pair

● **OOD** (OpenOOD protocol)

- near: CIFAR-100
- far: SVHN, DTD, MNIST, F-MNIST, Gaussian, Uniform

CIFAR-10H: the interval widens where humans disagree



Mean $[\mathrm{Bel}, \mathrm{Pl}]$ on the true class, 10,000 images, binned by annotator entropy.

Yager width: 0.008 $\rightarrow$ 0.043, monotone ($5\times$)

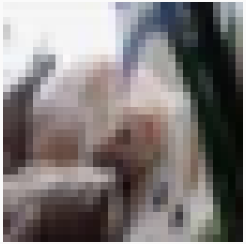
Dempster width: 0.002 $\rightarrow$ 0.005

	Yager	Dempster
Spearman ρ	0.278	0.263
Decile AUROC	0.77	0.76
$m(\Theta)$, top decile	**0.044**	0.005

Single seed.

Same ranking, $9\times$ the scale. Only Yager's $m(\Theta)$ is readable on its own $[0, 1]$ scale.

Two contested images



CIFAR-10 test #3799 (label: cat)
humans: cat 0.49, dog 0.30

● Yager $m(\Theta) = 0.126$
Dempster $m(\Theta) = 0.002$



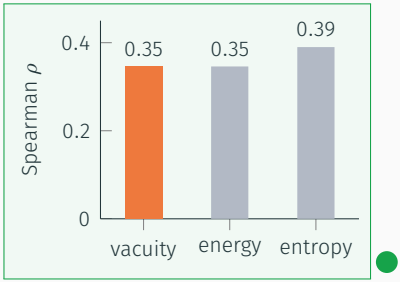
CIFAR-10 test #7524 (label: cat)
humans: ship 0.45, airplane 0.35

● Yager $m(\Theta) = 0.196$
Dempster $m(\Theta) = 0.015$

● For scale: mean Yager vacuity on the least-contested bin is 0.008.

ChaosNLI: the same signal in language

- 1,514 NLI pairs, 100 annotators each
- ● Vacuity vs. disagreement: $\rho = 0.346$, $p < 10^{-40}$
- ● Most vs. least contested quartile: **0.076** vs. **0.036**



● Vacuity does not *beat* the scalars. The point: **one quantity** tracks both modalities and comes with a $[\mathrm{Bel}, \mathrm{Pl}]$ interval.

OOD detection: a competitive by-product

Method	ID acc. (%)	Macro AUROC	Macro FPR95
Softmax + MSP*	94.33	0.653	0.853
Softmax + Energy*	94.33	0.863	0.345
EDL	94.04	0.895	0.337
MP-ENN-Yager	94.33	0.860	0.342
MP-ENN-Dempster†	94.26	0.912	0.292
Deep Ensemble (5)	94.20	**0.954**	**0.164**

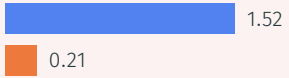
5 seeds, 7 OOD sets. Also: SNGP 0.886, MC Dropout 0.885, ODIN 0.871, DUQ 0.859.

* post-hoc scores on the Yager heads' logits (hence same accuracy).

† $1 - \kappa$ clamped.

What the clamp hides

Combination layer alone, clamp off, 200 random mass pairs on $\mathcal{F}$, $\bar{\kappa} = 0.75$:



mean $\|\nabla\|$: Dempster 7× Yager

Yager sits with energy scoring. Clamped Dempster is the best single-pass method, but needs the clamp. Yager needs no guard.

What did *not* work, and what barely matters

The learned discount α_s is inert

Head 2 fed Gaussian-corrupted features (σ=1.5, 3 seeds):
$\alpha = (0.99998,\ 0.99995,\ 0.99997)$. It *ranks* the corrupted head lowest every time, but a 3×10^{-5} discount does nothing. Shared encoder, same features: independent sources are the fair test.

Fusion order (non-associative)

- 3! orders $\rightarrow$ 3 distinct outputs
- CIFAR-10: ℓ_1 spread 0.002, no class changes
- CIFAR-100 ($\bar{\kappa} = 0.144$): 0.012; AUROC ± 0.0004

Ablations, cost

- $W \in \{1, 5, 10, 20, 100\}$: AUROC in $[0.855, 0.866]$
- $\lambda_{\text{unc}} \in \{0, 0.01, 0.1, 1\}$: $0.865 \rightarrow 0.850$
- Single pass, 51.9 ms (+14% over Energy)

Limitations and next steps

Limitations

- Quasi-Bayesian: no mass on $\{c_i, c_j\}$
- Yager is not associative
- Loss proper given $m(\Theta)$, not jointly (Gneiting; Bengs et al.)
- Human-alignment numbers are single-seed
- α-discount inert on a shared encoder

Next

- Independent sources (sensors, modalities, separate encoders): a fair test for discounting
- Conformal prediction over $[\mathrm{Bel}, \mathrm{Pl}]$
- Richer focal sets, e.g. learned class pairs

Take-home

1. Dempster's $1/(1-\kappa)$ puts $(1-\kappa)^{-2}$ in the gradient: it blows up exactly where sources disagree.
2. Yager's $\boxplus$ is bilinear and **1-Lipschitz**: the epistemic choice of 1987 is the one SGD needs.
3. With it, Shafer's mass and discount, Yager's combination and Jøsang's vacuity run **exactly** in one forward pass, and the output $m(\Theta)$ orders human disagreement on images and language **without being trained to**.

`github.com/sulcantonin/enn-public`

Thank you.

Backup slides

Backup: Zadeh's example, both rules

Two doctors: $m_1 = \{A{:}0.99,\ B{:}0.01\}$, $m_2 = \{C{:}0.99,\ B{:}0.01\}$. Conflict $\kappa = 1 - 0.01^2 = 0.9999$.

	A	B	C	Θ
Dempster $\oplus$	0	**1.0000**	0	0
Yager $\boxplus$	0	0.0001	0	**0.9999**

Dempster turns near-total disagreement into certainty about the option both doctors thought unlikely. Yager reports that the sources contradict each other. In a network, Dempster's gradient here is of order $(1-\kappa)^{-1} = 10^4$.

Backup: proof sketch of Theorem 1

Fix m_2. For $B \in \mathcal{F}$:

- $B = \{c_i\}$: $m_1(c_i)$ contributes $m_2(c_i) + m_2(\Theta)$ to $\{c_i\}$ and $\sum_{j \neq i} m_2(c_j)$ to Θ (conflict).
- $B = \Theta$: $m_1(\Theta)$ contributes $m_2(c_j)$ to each $\{c_j\}$ and $m_2(\Theta)$ to Θ.

Every entry lies in $[0, 1]$ and each column sums to $\sum_{C \in \mathcal{F}} m_2(C) = 1$: the Jacobian is column-stochastic, so $\|J\|_1 = 1$.

Fold of $|S|$ sources. A product of column-stochastic matrices is column-stochastic, so $\tilde{m}_s \mapsto m$ is 1-Lipschitz for every s, with the other sources fixed.

Dempster. Every entry is divided by $1 - \kappa(m_1, m_2)$, which depends on m_1: the map is no longer linear, and the quotient rule produces the $(1 - \kappa)^{-2}$ term.

Backup: relation to prior evidential networks

- **Denœux** (evidential k-NN 1995, neural classifier 2000, logistic regression as a DS layer 2019), **Tong et al.** 2021: belief-function layers on learned features, *combined by Dempster*. Our question: what happens to that choice once it is differentiated.
- **EDL** (Sensoy 2018), **PostNet**: Dirichlet outputs, DST at a single layer, no multi-source combination, no $[\mathrm{Bel}, \mathrm{Pl}]$ interval. Our Eq. 1 is EDL's parametrisation read as a mass.
- **Credal / random-set networks** (Manchingal, Wang, Caprio): set-valued prediction by other means.
- **Neurosymbolic** (LNN): real-valued surrogates for Bel/Pl that weaken duality; we keep it exact.

To our knowledge, the first direct comparison of Yager and Dempster as the combination layer of a back-propagated network.

Backup: likely questions (1), the rule

Other unnormalised rules are also 1-Lipschitz (Smets' conjunctive rule, Dubois–Prade). Yes: any bilinear rule without the normaliser is. Smets puts κ on $\emptyset$, which leaves the Shafer mass functions and needs a decision-time normalisation; Dubois–Prade puts it on unions $\{c_i, c_j\}$, which leaves $\mathcal{F}$. Yager is the one that stays a mass function on $\mathcal{F}$ with κ readable as ignorance. PCR rules reintroduce input-dependent denominators.

Did Dempster actually blow up in training? No. With the clamp at 10^{-8} it trains to 94.26%; in the 60-epoch ablation run its mean per-epoch gradient norm was even lower than Yager's (1.44 vs. 2.04, pre-clip). The blow-up is at the combination layer with the clamp off ($7\times$ on random mass pairs; $1/(1-\kappa)$ on the two-head family). The claim is about what the clamp costs: a guard with no DST reading that silences the high-conflict regime.

Why not clip gradients and keep Dempster? We clip (1.0, all methods; norms reported pre-clip). Clipping bounds the update but cannot fix a division by $1 - \kappa \to 0$: Dempster still needs the clamp.

Backup: likely questions (2), the human-alignment result

How does softmax entropy do on CIFAR-10H? (post-CRC check on the paper's Yager checkpoint, single seed, not in the paper.) Post-hoc on the same heads' logits: softmax entropy $\rho = 0.15$ (decile AUROC 0.57), energy $\rho = 0.27$ (0.76), vacuity $\rho = 0.28$ (0.77). On ChaosNLI entropy is slightly higher (0.39 vs. 0.35).

$\rho = 0.28$ **is small.** $n = 10{,}000$, $p \approx 10^{-176}$; 44% of the images are unanimous (entropy 0), which caps any rank correlation. The decile AUROC of 0.77 and the 9× scale are the readable numbers.

Is $m(\Theta)$ a rescaled entropy? No: it is the exact $[\mathrm{Bel}, \mathrm{Pl}]$ width, it decomposes into "nobody committed" $+ \kappa$, and it is monotone in annotator entropy without being trained on it.

Dempster ranks as well; why prefer Yager? Scale: 0.044 vs. 0.005 on the contested decile. Dempster's vacuity would need recalibration to be read as an amount of ignorance; Yager's is readable as is.

Backup: likely questions (3), design choices

EDL beats Yager on OOD. Yes (0.895 vs. 0.860); Yager matches energy scoring, and OOD is secondary here. Calibration: ECE 0.036 (Yager) vs. 0.050 (EDL).

Why not average over all fusion orders? Implemented (symmetric variant); a measured spread of 0.002 in ℓ_1 does not justify $|S|!$ cost.

More than 3 sources? $|S| = 3$ throughout; how vacuity grows with $|S|$ was not studied.

Why $W = K$? A uniform prior pseudo-count per class; results are flat over $W \in \{1, 5, 10, 20, 100\}$.

Why hard labels at all? To test the structural claim: if $m(\Theta)$ tracks disagreement without ever seeing it, the signal comes from the combination rule, not from supervision. Training on soft labels is a different, easier question.